

Vector-space containers (VCont)

A self-contained treatment: the biproduct collapse, its exact finite-field defect, algebraic structure, and the admissibility frontier

Author: MacBeth (Kodamai)

Date: 2026-09-05, rev. 2026-09-11

For review by: Rick

Corresponds to: scotmacbeth/work-in-progress commit bfcfd55

Draft — work in progress; not yet neighbour-reviewed. Subject matter prepared for Neil Ghani / an external requester.

Reading guide and honesty note. Every theorem below is tagged in prose with its epistemic grade: **proved** (a complete pen-and-paper proof exists in the cited internal file), **lean-verified** (machine-checked, sorry-free, in Lean 4), or **OPEN** (a precisely stated conjecture that is *not* settled). The headline result of §2 is proved and lean-verified. The frontier question of §4 (“Question (Q)”) is **OPEN** and is conjectured independent of ZFC; it is presented as such and nowhere treated as settled. This note has not yet had a neighbour review.

Fix a field k . A *container* in a category $\mathcal{C}$ is a pair (shapes, positions): a set S together with a position object $P_s \in \mathcal{C}$ for each $s \in S$, extending to an endofunctor $\llbracket S, P \rrbracket$ of $\mathcal{C}$. Over $\mathcal{C} = \mathbf{Set}$ this is the classical theory of Abbott, Altenkirch and Ghani: containers are precisely the polynomial functors, and the extension functor is a full and faithful embedding. This note asks what survives when the positions are *vector spaces* — the base $\mathcal{C} = \mathbf{Vec}_{\text{fd}}$ of finite-dimensional k -vector spaces — and reports the answers the programme has established. The one-line summary: *the coproduct/product distinction that makes Set containers rigid becomes a biproduct over Vec, and much of what the Set theory reads off “for free” collapses.* Over a finite field $\mathbb{F}_q$ that collapse is not merely qualitative: §3 computes its exact size $\delta = \prod_s a_s - \prod_s (a_s - (|T| - 1))$ as a closed integer point-count.

1 Definition of VCont and its extension functor

Write $\mathbf{Vec} = \mathbf{Vec}_k$ for all k -vector spaces, $\mathbf{Vec}_{\text{fd}}$ for the finite-dimensional ones, and $\mathbf{Vec}(P, W)$ for the hom vector space. All functors below are **Vec**-functors: their action on homs $\mathbf{Vec}(V, W) \rightarrow \mathbf{Vec}(FV, FW)$ is k -linear.

Definition 1 (linear / vector-space container). A *vector-space container* (object of $\mathbf{VCont} := \text{Fam}(\mathbf{Vec}^{\text{op}})$, or of $\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$ when finiteness is imposed) is a pair $(S, (P_s)_{s \in S})$ with S a set (the *shapes*) and each $P_s \in \mathbf{Vec}$ a *position space*. A morphism $(S, (P_s)) \rightarrow (T, (Q_t))$ is a pair $(f, (\varphi_s))$ with $f: S \rightarrow T$ a function and, for each $s \in S$, a linear map $\varphi_s: Q_{f(s)} \rightarrow P_s$ — *covariant on shapes, contravariant on positions*, exactly as for **Set** containers. Write $n_s := \dim P_s$.

Definition 2 (extension). The *extension* of (S, P) is the functor $\llbracket S, P \rrbracket: \mathbf{Vec} \rightarrow \mathbf{Vec}$,

$$\llbracket S, P \rrbracket W := \bigoplus_{s \in S} \mathbf{Vec}(P_s, W).$$

Choosing a basis of P_s gives $\mathbf{Vec}(P_s, W) \cong W^{n_s}$ naturally in W , so $\llbracket S, P \rrbracket W \cong \bigoplus_{s \in S} W^{\oplus n_s}$, a “direct sum of powers”. On a morphism $(f, (\varphi_s))$ the natural transformation is $(g_s)_s \mapsto (g_{f(s)} \circ \varphi_s)_s$.

Writing $h_P := \mathbf{Vec}(P, -)$ for the corepresentable at P , we have $\llbracket S, P \rrbracket = \bigoplus_{s \in S} h_{P_s}$ and $h_k = \mathbf{Vec}(k, -) \cong \text{Id}$.

Two features of **Vec** shape everything that follows.

Coproduct = product = biproduct. A finite coproduct in **Vec** is a biproduct, hence *equals* the product. Thus although the outer sum $\bigoplus_{s \in S}$ in Definition 2 is written as a coproduct (a “ Σ over shapes” of the hom-spaces $\mathbf{Vec}(P_s, X)$), for finite S its underlying space is equally the product $\prod_{s \in S} \mathbf{Vec}(P_s, X)$ — a “ Π over shapes”. This $\Sigma \rightsquigarrow \Pi$ coincidence is the *biproduct collapse*, and it is the single change from the set-level extension $\llbracket S, P \rrbracket X = \prod_{s \in S} X^{P_s}$ (a genuine disjoint union). It is made fully explicit in the Lean formalisation of §6, where the linear extension is literally $\mathbf{LExt}(S, \text{Pos}) X = (s : S) \rightarrow (\text{Pos } s \rightarrow X)$.

Finite positions distribute over $\oplus$. If $\dim P < \infty$ then $\mathbf{Vec}(P, \bigoplus_t V_t) \cong \bigoplus_t \mathbf{Vec}(P, V_t)$ naturally; equivalently, h_P preserves coproducts iff P is finite-dimensional (a finite-support argument on a basis of P). This is why we usually restrict positions to **Vec**_{fd}; the infinite-dimensional case is the source of the frontier in §5.

Example 1. Take $S = \{a, b\}$ with $P_a = k^2$, $P_b = k$. Then $\llbracket S, P \rrbracket W = \mathbf{Vec}(k^2, W) \oplus \mathbf{Vec}(k, W) \cong W^2 \oplus W \cong W^3$, so $\llbracket S, P \rrbracket \cong \text{Id}^3$. Numerically $\dim \llbracket S, P \rrbracket(k^d) = 3d$ for every d (verified computationally, $d = 0, \dots, 4$). The two integers that produced this functor — the shape S and the partition $3 = 2 + 1$ — have already vanished from the answer: only the total $N = \sum_s n_s = 3$ remains. That disappearance is the subject of §2.

2 The collapse: $\llbracket - \rrbracket$ is neither full nor faithful over **Vec**, and not injective on objects

Over **Set** the extension functor $\llbracket - \rrbracket : \mathbf{Fam}(\mathbf{Set}^{\text{op}}) \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is **full and faithful** (Abbott–Altenkirch–Ghani): the container data (S, P) can be recovered from the functor, with $S = \llbracket S, P \rrbracket(1)$ the value at a one-element set. Over **Vec** this recovery fails at its root.

Remark 1 (the terminal slogan dies). The terminal object of **Vec** is the zero object 0 , so $\llbracket S, P \rrbracket(0) = \bigoplus_s \mathbf{Vec}(P_s, 0) = 0$: the shapes are invisible at the terminal object, and the **Set** slogan “ $F(1) = S$ ” has no analogue.

2.1 The finite collapse (headline; proved and lean-verified)

Theorem 1 (finite collapse; **proved**). *Let S be finite and every $n_s = \dim P_s$ finite, and put $N := \sum_{s \in S} n_s = \dim(\bigoplus_s P_s)$. Then there is a natural isomorphism $\llbracket S, P \rrbracket \cong \text{Id}^N$. Consequently:*

1. $\llbracket S, P \rrbracket \cong \llbracket S', P' \rrbracket$ as functors **iff** $N = N'$; a finite vector-space container is classified up to isomorphism of its extension by the single number N .
2. Since $\text{End}(\text{Id}) = k$ is local and $\text{End}(\text{Id}^N) = M_N(k)$ is semiperfect, Krull–Schmidt–Azumaya makes the decomposition into indecomposables unique; the recovered invariant is exactly the multiplicity N . The shape set S and the partition $N = \sum_s n_s$ are not recoverable.

This is proved in `proofs/2026-08-18-linear-containers-vec.md` (Theorem 2.1; registry `linear-containers-vec.json`, node `part1-finite-collapse`, grade **proved**), with the co-Yoneda lemma $\text{Nat}(h_P, G) \cong G(P)$ and the additivity of $P \mapsto h_P$ as the engine.

Theorem 2 (the explicit witness; **proved, lean-verified**). *The two containers*

$$p = (\{*\}, k^2) \quad \text{and} \quad p' = (\{1, 2\}, (k, k))$$

*both have extension Id^2 , i.e. both present the functor $X \mapsto X \oplus X$, yet they are **not isomorphic as containers**: their shape sets have different cardinalities ($1 \neq 2$). Hence $\llbracket - \rrbracket$ is not injective on objects.*

Contrast **Set**: there the analogous containers $(\{*\}, 2)$ and $(\{1, 2\}, (1, 1))$ have *distinct* extensions X^2 and $2 \cdot X$, separated already at $X = 1$. The collapse is precisely the failure of that separation. Theorem 2 is machine-checked; see §6.

2.2 Why: the hom formula and the failure of extensivity

The mechanism is visible one categorical level up, on hom-sets. Both computations below are proved in 2026-08-18-linear-containers-vec.md (Theorems 3.1–3.3; registry nodes **part2-hom-formula**, **part2-extensivity-crux**, both **proved**).

Proposition 1 (hom formula; **proved**). *For vector-space containers (S, P) , (T, Q) ,*

$$\begin{aligned} \text{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) &\cong \prod_{s \in S} \bigoplus_{t \in T} \mathbf{Vec}(Q_t, P_s), \\ \text{Fam}(\mathbf{Vec}^{\text{op}})((S, P), (T, Q)) &\cong \prod_{s \in S} \prod_{t \in T} \mathbf{Vec}(Q_t, P_s), \end{aligned}$$

*where $\coprod$ is the coproduct in **Set** (disjoint union): a container morphism assigns to each s a single choice $t = f(s)$ together with a linear map in $\mathbf{Vec}(Q_{f(s)}, P_s)$.*

Theorem 3 (neither full nor faithful; **proved**). *The functor $\llbracket - \rrbracket$ induces on each hom the canonical comparison*

$$\prod_s \prod_t \mathbf{Vec}(Q_t, P_s) \longrightarrow \prod_s \bigoplus_t \mathbf{Vec}(Q_t, P_s),$$

*componentwise the map from the disjoint union $\coprod_t$ into the biproduct $\bigoplus_t$ sending a choice (t, φ) to the element supported at t . Over $\mathbf{Vec}$ this map is neither surjective nor injective, so $\llbracket - \rrbracket$ is **neither full nor faithful**.*

1. Not full as soon as some shape s admits two shapes $t \neq t'$ with $\mathbf{Vec}(Q_t, P_s) \neq 0 \neq \mathbf{Vec}(Q_{t'}, P_s)$: any natural transformation whose s -component is a genuine linear combination $\varphi_t + \varphi_{t'}$ supported on two shapes is not in the image of a shape-function.
2. Not faithful as soon as $S \neq \emptyset$ and $|T| \geq 2$. Over a field the zero linear map lies in every $\mathbf{Vec}(Q_t, P_s)$, so the two distinct container morphisms $(u, 0)$ and $(u', 0)$ — with $u \neq u' : S \rightarrow T$ and all position maps zero — are collapsed to the same (zero) natural transformation. The mechanism is exactly the biproduct: $\bigoplus_t$ has a single shared zero across all summands, so it merges the $|T|$ distinct disjoint-union-tagged zeros $(t, 0)$ into that one zero element; the comparison is injective only away from the zero maps. Hence $\llbracket - \rrbracket$ is **faithful iff** $|T| \leq 1$ **or** $S = \emptyset$. By contrast **Set**'s coproduct is the disjoint union, which keeps a distinct tag on each zero, so the classical Abbott–Altenkirch–Ghani $\llbracket - \rrbracket$ over **Set** remains faithful.

Remark 2 (provenance of the faithfulness correction). The failure of faithfulness was surfaced by R. Langer's finite-field analogue (q-container Cor. 5.3, Ex. 5.4); over $\mathbb{F}_q$ it is a finite point-count, made exact in §3. (Draft note.)

The point is a single symbol. Over **Set** the same computation gives $\text{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) \cong \prod_s \bigoplus_t \mathbf{Set}(Q_t, P_s)$ where now $\bigoplus_t = \coprod_t$ is the coproduct *in Set*, i.e. the disjoint union: container-hom equals Nat, and $\llbracket - \rrbracket$ is full and faithful. **Set is extensive** — its coproduct is the disjoint union — and that is exactly what makes $\llbracket - \rrbracket$ full. **Vec is not extensive** — its coproduct is a biproduct — and the fullness gap

$$\coprod_t \subsetneq \bigoplus_t$$

is the precise, quantified manifestation of that failure. The object-level collapse (Theorem 1) and the morphism-level fullness gap (Theorem 3) are *one* phenomenon: the failure of extensivity under the base change $\mathbf{Set} \rightsquigarrow \mathbf{Vec}$.

Remark 3 (what survives; **proved**). The collapse used both finiteness of S and finite-dimensionality of positions; drop either and content returns. With infinite S and all $P_s = k$, $\llbracket S, P \rrbracket = \text{Id}^{\oplus |S|}$ is a genuine coproduct, not a product, of copies of Id ; and for a single infinite-dimensional P , $h_P = \mathbf{Vec}(P, -)$ is an infinite-product functor, not built from Id under $\oplus$, and P is recovered from the representable (Lemma 1.2/1.3 of the source file). The finite recovery is sharp: $\llbracket - \rrbracket$ determines the total space $\bigoplus_s P_s$ up to isomorphism, and nothing about its partition into shapes (Corollary 3.4, **proved**).

3 The finite-field instance: q -containers and the exact collapse defect

Everything in §2 is stated over an arbitrary field k and is *qualitative*: $\llbracket - \rrbracket$ either is or is not full, is or is not faithful. Over a *finite* field $k = \mathbb{F}_q$ these failures become finite point-counts, and the size of the collapse can be written in closed form. This is the setting of R. Langer’s q -container note (personal communication, 2026-09-11): Definition 1 verbatim, with positions finite-dimensional $\mathbb{F}_q$ -spaces. His §4 morphism count and §5 faithfulness corollary are the first two displays below; the exact *defect* (Theorem 4) is a joint refinement. This subsection is thus the arithmetic shadow of the whole of §2: the one structural symbol $\coprod_t \rightsquigarrow \bigoplus_t$ becomes the one integer $-(|T| - 1)$.

Fix $\mathbb{F}_q$ and q -containers $C = (S, (P_s))$, $D = (T, (Q_t))$ with S, T finite; write $d_s = \dim P_s$, $e_t = \dim Q_t$, and $\mathbf{Cont}_q(C, D) := \text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})((S, P), (T, Q))$ for the (now finite) morphism set. The extension map on this hom, $\llbracket - \rrbracket : \mathbf{Cont}_q(C, D) \rightarrow \text{Nat}(\llbracket C \rrbracket, \llbracket D \rrbracket)$, has finite source and target, and Proposition 1 specialises to two counts:

$$|\mathbf{Cont}_q(C, D)| = \prod_{s \in S} \left(\sum_{t \in T} q^{e_t d_s} \right), \quad |\text{Nat}(\llbracket C \rrbracket, \llbracket D \rrbracket)| = q^{(\sum_s d_s)(\sum_t e_t)}. \quad (1)$$

The first is a product of independent per-shape choices: at each s pick a target $u(s) = t$ and a map in $\mathbf{Vec}(Q_t, P_s)$, of which there are $q^{e_t d_s}$. The second is co-Yoneda: with $P := \bigoplus_s P_s$ and $Q := \bigoplus_t Q_t$ the extension is corepresentable, $\llbracket C \rrbracket = \mathbf{Vec}(P, -)$, so $\text{Nat}(\llbracket C \rrbracket, \llbracket D \rrbracket) \cong \mathbf{Vec}(Q, P)$, of cardinality $q^{\dim Q \cdot \dim P}$.

Under that co-Yoneda bijection a morphism $(u, (f_s))$ becomes the block matrix $M \in \mathbf{Vec}(Q, P)$, rows indexed by s and columns by t , whose s -th row carries the single block f_s in column $u(s)$ and zero elsewhere. So as (u, f) ranges over $\mathbf{Cont}_q(C, D)$ the induced matrix is built *row by row*, *independently across rows*, and each row g_s ranges over the set of tuples supported in a single column:

$$A_s = \bigcup_{t \in T} W_t^{(s)}, \quad W_t^{(s)} := \{ g \in \bigoplus_{t'} \mathbf{Vec}(Q_{t'}, P_s) : g_{t'} = 0 \ (t' \neq t) \} \cong \mathbf{Vec}(Q_t, P_s).$$

The image of $\llbracket - \rrbracket$ is therefore the product $\prod_s A_s$, and the only nontrivial count is $|A_s|$: a union of $|T|$ coordinate subspaces of a direct sum, which pairwise meet only in 0.

Lemma 1 (union of pairwise-trivially-meeting subspaces; **proved**). *If $W_1, \dots, W_n$ ($n \geq 1$) are finite subspaces of an $\mathbb{F}_q$ -space with $W_i \cap W_j = \{0\}$ for $i \neq j$, then $|\bigcup_i W_i| = (\sum_i |W_i|) - (n - 1)$.*

Each W_i contains 0; for $i \neq j$ the only shared vector is 0, so the punctured sets $W_i \setminus \{0\}$ are pairwise disjoint and $|\bigcup_i W_i| = 1 + \sum_i (|W_i| - 1) = (\sum_i |W_i|) - (n - 1)$. Applied to the $|T|$ coordinate subspaces $W_t^{(s)}$, with $|W_t^{(s)}| = q^{e_t d_s}$, this gives (for $|T| \geq 1$) $|A_s| = (\sum_t q^{e_t d_s}) - (|T| - 1)$.

Theorem 4 (collapse-defect formula; **proved**). *Write $a_s := \sum_{t \in T} q^{e_t d_s}$. For q -containers $C = (S, (P_s))$, $D = (T, (Q_t))$ with $|T| \geq 1$,*

$$|\text{image } \llbracket - \rrbracket| = \prod_{s \in S} (a_s - (|T| - 1)), \quad \delta(C, D) := |\mathbf{Cont}_q(C, D)| - |\text{image } \llbracket - \rrbracket| = \prod_s a_s - \prod_s (a_s - (|T| - 1)).$$

Combine $|\text{image } \llbracket - \rrbracket| = \prod_s |A_s|$ with Lemma 1 and subtract (1). Proved in `proofs/2026-09-11-collapse-defect-formula.md` (Theorem 5; registry `linear-containers-vec.json`, node `collapse-defect-formula`, grade **proved**), brute-force verified on 15 cases over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4$ (dims ≤ 2 , $|S|, |T| \leq 3$; `scratch/collapse_defect_verify.py`), all exact.

The formula makes the two qualitative statements of §2 into arithmetic:

Corollary 1 (faithful/full, quantified; **proved**). *For $|T| \geq 1$: $\llbracket - \rrbracket$ is faithful $\iff \delta(C, D) = 0 \iff |T| = 1$ or $S = \emptyset$ (the threshold of Theorem 3, now read off as “every factor $a_s - (|T| - 1)$ equals a_s ”); and $\llbracket - \rrbracket$ is full $\iff$ every factor A_s fills its space $\iff$ for each shape s with $P_s \neq 0$ at most one target t has $Q_t \neq 0$.*

The single term $-(|T| - 1)$ is the whole story. It counts exactly the $|T|$ distinct disjoint-union tags $(t, 0)$ that the biproduct $\bigoplus_t$ merges into its one shared zero: the $|T|$ morphisms $(u, f=0)$ with different shape-functions $u: S \rightarrow T$ all induce the same zero row. Over **Set** the corresponding union is a genuine disjoint union $\coprod_t$, which keeps a distinct tag on every zero; the correction term vanishes, $\delta = 0$, and the Abbott–Altenkirch–Ghani representation is full and faithful. Thus the finite-field defect is the failure of extensivity (§2) rendered as an integer.

Remark 4 (the $q \rightarrow 1$ limit does *not* recover **Set**-faithfulness; **proved**). It is tempting to read δ through a “field with one element”. Formally $a_s \rightarrow |T|$ and $a_s - (|T| - 1) \rightarrow 1$, so $|\text{image}| \rightarrow 1$ while $|\mathbf{Cont}| \rightarrow |T|^{|S|}$: the $q \rightarrow 1$ degeneration is *maximal* collapse, $\delta \rightarrow |T|^{|S|} - 1 \neq 0$, not the faithful **Set** case. The reason is structural: the **Set** extension $V \mapsto \coprod_s \mathbf{Set}(P_s, V)$ is a *different functor* (a coproduct of representables), full and faithful by AAG; the linear extension replaces that **Set**-coproduct by the biproduct. The $q \rightarrow 1$ limit reflects only that every hom-space collapses over $\mathbb{F}_1$, not any set-level faithfulness. The two functors must not be conflated.

The witness that surfaced the whole faithfulness correction is the smallest nontrivial instance: $C = (\{*\}, k)$, $D = (\{1, 2\}, (k, k))$ over $\mathbb{F}_2$ gives $|\mathbf{Cont}| = 4$, $|\text{image}| = 3$, $\delta = 1$ — two of the four morphisms (the two zero maps with $u = 1$ vs. $u = 2$) collapse to one natural transformation (Langer, Ex. 5.4). A sample of the verified table:

q	C	D	$ \mathbf{Cont} $	$ \text{image} $	$ \text{Nat} $	δ
2	(1)	(1, 1)	4	3	4	1
2	(1, 1)	(1, 1)	16	9	16	7
2	(1)	(1, 1, 1)	6	4	8	2
2	(2)	(1, 1)	8	7	16	1
3	(1, 1)	(1, 1)	36	25	81	11
2	(1, 1)	(2)	16	16	16	0 ($ T =1$)

Here C, D are written as the tuples of position dimensions $(d_s), (e_t)$. The $|T| = 1$ row realises Corollary 1's faithful case ($\delta = 0$); the $(2) \rightarrow (1, 1)$ row is the Theorem 2 witness direction. The finite-dimensional theory over a finite field is thus not merely qualitatively but *numerically* pinned down.

4 Algebraic structure

Composition of vector-space containers is available on the finite-dimensional locus, and it too is shaped by the collapse.

Proposition 2 (composition; **proved**). *For containers with finite-dimensional positions,*

$$\llbracket S, P \rrbracket \circ \llbracket T, Q \rrbracket \cong \llbracket S \times T, (P_s \otimes Q_t)_{(s,t)} \rrbracket,$$

so on $\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$ the composition product is $(S, P) \triangleleft (T, Q) = (S \times T, (P_s \otimes Q_t))$, with unit $I = (\{*\}, k)$ (extension Id).

This is *not* the **Set** container composition, whose shapes form the dependent sum $\coprod_{s \in S} T^{P_s}$: linearity flattens the dependency (a linear map $P_s \rightarrow \bigoplus_t V_t$ is a *sum of components*, not a *choice of branch*) into the plain product $S \times T$. Proved in 2026-08-18-linear-containers-vec.md, Prop. 4.1 (registry node **part3-composition**, **proved**).

The comonoids for this product identify a clean algebraic species.

Theorem 5 ($\triangleleft$ -comonoids are families of algebras; **proved**). *Let $C = (S, P) \in \text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$ with S an arbitrary set and each P_s finite-dimensional. The $\triangleleft$ -comonoid structures (δ, ε) on C in $(\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}}), \triangleleft, I)$ are in natural bijection with families $(A_s)_{s \in S}$ of unital associative k -algebra structures $A_s = (P_s, \mu_s, \eta_s)$, one on each position space. The bijection is $\delta_{\text{shape}} = \Delta$ (the diagonal), $\delta_s^\sharp = \mu_s$, $\varepsilon_s^\sharp = \eta_s$. On morphisms this upgrades to an isomorphism of categories*

$$\text{Comon}_{\triangleleft}(\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})) \cong \text{Fam}(\mathbf{Alg}_k^{\text{op}}),$$

the cocommutative comonoids forming the full subcategory $\text{Fam}(\mathbf{CAlg}_k^{\text{op}})$.

Proved in proofs/2026-08-19-vec-comonoids-algebras.md (registry node **part3-comonoid-algebras**, grade **proved**; hypotheses sharpened to finite-dimensional positions only, S arbitrary). The counit forces the shape comultiplication to the diagonal (the unique $(\mathbf{Set}, \times)$ -comonoid on S); contravariance turns $\delta_s^\sharp$ into a multiplication $P_s \otimes P_s \rightarrow P_s$ and $\varepsilon_s^\sharp$ into a unit $k \rightarrow P_s$; the comonoid laws become associativity and the two-sided unit law.

Remark 5 (the algebroid guess, refuted in finite dimension; **proved**). Over **Set** the analogous fact is $\triangleleft$ -comonoid $\cong$ small category (Ahman–Uustalu; **Cont** $\cong$ Fam(**Set**^{op})). One might hope $\triangleleft$ -comonoid over **Vec** $\cong$ k -linear category (algebroid, Mitchell). This is *false* in finite dimension: because the counit forces $\delta_{\text{shape}} = \Delta$, the comultiplication never reaches an off-diagonal block $P_a \otimes P_b$ ($a \neq b$), so there is no hom-between-distinct-objects composition. The structure is a disjoint family of one-object k -linear categories = k -algebras. A genuine algebroid requires double-indexed positions and a matrix (bimodule) composition $(P \circ Q)_{a,c} = \bigoplus_b P_{a,b} \otimes Q_{b,c}$ — the $\bigoplus_b$ restoring the extensivity coproduct whose absence caused the strict collapse. That identification is classical (Bénabou; Mitchell) and is treated separately; the finite single-index result stands as above.

Which monoidal / closed structures survive (high level)

Three companion results from the wider “containers over a base” programme locate the collapse inside the closed-structure theory. They are stated here at high level and *not* re-proved; each is proved in its own source file.

- **T1 (unit connectedness; proved)**. For a closed symmetric monoidal cocomplete base $\mathcal{C}$, the extension functor $\llbracket - \rrbracket : \text{Fam}(\mathcal{C}^{\text{op}}) \rightarrow [\mathcal{C}, \mathcal{C}]$ is full and faithful *iff* the monoidal unit is *connected* ($\mathcal{C}(I, -)$ preserves coproducts). Over **Vec** the unit is *disconnected* (the zero object makes $|-| : \mathbf{Vec}_{\text{fd}} \rightarrow \mathbf{Set}$ fail to preserve coproducts), which is exactly why $\llbracket - \rrbracket$ is not full — a base-general explanation of §2.
- **T2 (Dirichlet / Day tensor; proved)**. The pointwise Dirichlet tensor on Fam($\mathcal{C}^{\text{op}}$) is Day convolution, and its closedness is controlled by a familial-representability condition that fails over both **Vec** and **Vec**_{fd}.
- **T4-left (left closure; proved)**. The composition product $\triangleleft$ is left-closed on the copower-tiny (collapse) locus, where $p \triangleleft q = p \otimes q$; this is the locus on which Proposition 2 lives.

(These are internal results of MacBeth’s programme; grades are as recorded in memory and are cited, not re-derived, here.)

5 Composition and admissibility

Proposition 2 defines $\triangleleft$ on *finite-dimensional* positions. The general question is whether Fam(**Vec**^{op}) carries a composition product at all.

Definition 3 ($\triangleleft$ -admissibility). Fam($\mathcal{C}^{\text{op}}$) is $\triangleleft$ -*admissible* if the image of $\llbracket - \rrbracket$ in $[\mathcal{C}, \mathcal{C}]$ is closed under composition: for all p, q there is r with $\llbracket r \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$.

Proposition 3 (admissibility is per-object; **proved**). Call $P \in \mathcal{C}$ absorptive if $X \mapsto [P, \llbracket q \rrbracket X]$ is an extension for every q . Then Fam($\mathcal{C}^{\text{op}}$) is $\triangleleft$ -admissible *iff* every object of $\mathcal{C}$ is absorptive. There are two sources of absorptivity: copower-tiny (flexible) objects, where $[P, \coprod_t Y_t] = \coprod_t [P, Y_t]$; and distributive/extensive (rigid) objects, via the **Set**-style mechanism.

Proposition 6.1 of proofs/2026-09-01-gap1-setxvec-proved.md (**proved**). In **Vec**_{fd} every object is copower-tiny (finite-dimensional = finitely generated = copower-tiny; Lemma 5.1 there, **proved**), so Fam(**Vec**_{fd}^{op}) is admissible on the collapse locus with $\triangleleft = \otimes_k$ — this is Proposition 2 again.

5.1 A proved positive result: $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is admissible

Theorem 6 (Gap 1 inhabited; **proved**). *On the finite-vec-support locus $\mathcal{P} \subseteq \text{Fam}((\mathbf{Set} \times \mathbf{Vec}_{\text{fd}})^{\text{op}})$ (shape S and set-positions A_s arbitrary; $\{s : V_s \neq 0\}$ finite), the base $\mathcal{C} = \mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is $\triangleleft$ -admissible: for all $p, q \in \mathcal{P}$ there is $r = p \triangleleft q \in \mathcal{P}$ and an explicit natural isomorphism $\Theta: \llbracket p \triangleleft q \rrbracket \Rightarrow \llbracket p \rrbracket \circ \llbracket q \rrbracket$ — both components built as maps (the $\mathbf{Set}$ -component a bijection with explicit inverse, the $\mathbf{Vec}$ -component a natural composite of hom-out-of-coproduct, tensor-hom, and finite-support isomorphisms), not matched merely by cardinality.*

Theorem 3.1 of `proofs/2026-09-01-gap1-setxvec-proved.md` (**proved**; the map was built and its bijectivity, invertibility, and naturality confirmed computationally with zero failures). The base $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is non-collapse, non-cartesian, and has a disconnected unit, so it exhibits admissibility *without* unit-connectedness: it factors a rigid (extensive) $\mathbf{Set}$ part against a flexible (copower-tiny) $\mathbf{Vec}$ part. One honesty caveat, itself proved: the product $\triangleleft$ carries *no canonical* monoidal structure — the construction must choose a concentration slot — and the obstruction to canonicity is exactly the non-injectivity of $\llbracket - \rrbracket$ on objects (Theorem 2). A non-canonical monoidal structure does exist.

5.2 The open frontier: is full $\mathbf{Vec}$ admissible? (Question (Q), OPEN)

For *finite-dimensional* positions everything works. For *full* $\mathbf{Vec}$ the issue is that an infinite-dimensional position P is not copower-tiny, so $[P, \bigoplus_t (-)]$ need not be an extension. By Proposition 3 the whole question reduces to a single test object. The sharpest one is $E := k^{(\mathbb{N})}$ (countable-dimensional, free, non-tiny) against $q = (\mathbb{N}, (k))$, giving $\llbracket q \rrbracket X = \bigoplus_{\mathbb{N}} X$ and

Question 1 (the frontier; **OPEN**). Is

$$F(X) := \text{Hom}(E, \bigoplus_{\mathbb{N}} X) = \prod_{\mathbb{N}} \left(\bigoplus_{\mathbb{N}} X \right)$$

(row-finite $\mathbb{N} \times \mathbb{N}$ matrices over X)

naturally isomorphic to a coproduct of representables $\bigoplus_j \text{Hom}(N_j, -)$ in the additive functor category $\text{Add}(\mathbf{Vec}, \mathbf{Vec})$? Equivalently: is F a *projective* object of $\text{Add}(\mathbf{Vec}, \mathbf{Vec})$?

The stakes: **(Q) YES** would give an irreducible admissible base beyond Theorem 6; **(Q) NO** would confirm the “absorptive dichotomy” conjecture (every nice admissible base decomposes as rigid-extensive $\times$ flexible-additive). *This is the one deep open problem in the theory and we state it as unsettled.*

What *is* proved is a precise homological reduction, all in ZFC, in `proofs/2026-09-02-vec-admissibility-rigidity.md` and `proofs/2026-09-04-Q-homological-reduction-and-target-dimension-dichotomy.md`:

- **Lemma R (Yoneda rigidity; proved)**. Every $\alpha \in \text{Nat}(F, \text{id})$ has $\alpha \circ \text{in}_n = 0$ for all but finitely many rows n . The engine is pure Yoneda: $\prod_{\mathbb{N}} = h_E$ and $\text{Nat}(h_E, \text{id}) = E = k^{(\mathbb{N})}$ is finite-support. This resurrects Baer–Specker-type rigidity *over a field*, where module slenderness fails.
- **Theorem E (homological reduction; proved, ZFC)**. With $F_{\text{fin}} := \bigoplus_n A \subseteq F$ ($A := \bigoplus_m \text{id}$) and $Q := F/F_{\text{fin}}$, for every M there is an exact sequence $0 \rightarrow \text{coker}(\rho_M) \rightarrow \text{Ext}^1(Q, M) \rightarrow \text{Ext}^1(F, M) \rightarrow 0$. Hence

$$F \text{ is projective} \iff \text{Ext}^1(F, M) = 0 \text{ for all } M.$$

So (Q) is exactly the single homological question “is $\text{Ext}^1(F, -) \equiv 0$?”.

- **Theorem H (cardinality is neutral; proved, ZFC).** The id-dual $D(h_N) = \text{Nat}(h_N, \text{id}) = N$ has *no* $2^{\dim}$ blow-up (unlike $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}^{(\kappa)}, \mathbb{Z})$), so no cardinal invariant $\dim F(X)$ or $\dim \text{Nat}(F, M)$ separates F from a free functor. Both classical decision routes — single-target rigidity (all such Nat -groups vanish on Q) and cardinality — are thereby *provably closed*; any resolution must engage $\text{Ext}^1(F, -)$ at an infinite-dimensional target.

Remark 6 (status of (Q); **OPEN**, conjectured independent of ZFC). Question 1 is **not settled**. Because both the cardinality engine (Theorem H) and the single-target engine are provably unavailable over a field, the residual is a Whitehead-type extension problem on the syzygies of F . The current *conjecture* — theorem-supported but *not* proved — is that (Q) is **independent of ZFC**: plausibly NO under $2^{\aleph_0} < 2^{\aleph_1}$ and possibly YES under a failure of that hypothesis plus a uniformisation. Settling (Q) requires genuine set-theoretic input (Shelah; Eklof–Mekler transported to functor categories) and is the flagged frontier of the programme.

6 Formalisation

The object-level collapse (Theorem 2) is machine-checked in Lean 4 core, with *no* Mathlib dependency, in `lean/Containers/Containers/VecCollapse.lean`. Positions are carried by their finite basis types (k^2 by `Bool`, k by `Unit`), and the linear extension is modelled as the Π -over-shapes $\text{LExt}(S, \text{Pos}) X = (s : S) \rightarrow (\text{Pos } s \rightarrow X)$ — the single $\Sigma \rightsquigarrow \Pi$ shift that *is* the biproduct collapse. The two containers $p = (\{*\}, k^2)$ and $p' = (\{1, 2\}, k)$ appear as `VecCollapse.p` and `VecCollapse.p'`. The theorem `Containers.VecCollapse.collapseIso` exhibits a full natural isomorphism `LinNatIso p p'` (both directions, both triangle identities, and naturality in X), while `Containers.VecCollapse.shapes_not_equiv` certifies (`sorry-free`, and axiom-free — `#print axioms` reports `[]`) that the shape sets `Unit` and `Bool` admit no bijection. Together they are the machine-checked witness that $\llbracket - \rrbracket$ presents the same functor $X \mapsto X \oplus X$ by two non-isomorphic containers — Theorem D(ii), **lean-verified**. The isomorphism `collapseIso` depends only on `Quot.sound` (function extensionality).

7 Summary and outlook

Statement	Source	Grade
Extension $\llbracket S, P \rrbracket W = \bigoplus_s \mathbf{Vec}(P_s, W); \Sigma \rightarrow \Pi$ biproduct form	2026-08-18, Def. 0.2	definition
Finite collapse $\llbracket S, P \rrbracket \cong \text{Id}^N$; classified by N alone (Thm 1)	2026-08-18, Thm 2.1	proved
Witness $(\{*\}, k^2) \not\cong (\{1, 2\}, k)$, same functor $X \mapsto X \oplus X$ (Thm 2)	2026-08-18 / VecCollapse.lean	proved, lean-verified
$\llbracket - \rrbracket$ neither full nor faithful; faithful iff $ T \leq 1$ or $S = \emptyset$ (Thm 3)	2026-08-18, Thm 3.3	proved
$\mathbb{F}_q$ defect $\delta = \prod_s a_s - \prod_s (a_s - (T - 1))$, $a_s = \sum_t q^{e_t d_s}$ (Thm 4)	2026-09-11, Thm 5	proved
Composition $(S, P) \triangleleft (T, Q) = (S \times T, (P_s \otimes Q_t))$ (Prop 2)	2026-08-18, Prop 4.1	proved
$\triangleleft$ -comonoids $\cong$ families of k -algebras (Thm 5)	2026-08-19, Thm 7	proved
Admissibility $\Leftrightarrow$ every object absorptive (Prop 3)	2026-09-01, Prop 6.1	proved
$\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is $\triangleleft$ -admissible (Thm 6)	2026-09-01, Thm 3.1	proved
Homological reduction: $F \text{ proj.} \Leftrightarrow \text{Ext}^1(F, -) \equiv 0$ (Thm E)	2026-09-04, Thm E	proved (ZFC)
No cardinal invariant separates F from free (Thm H)	2026-09-04, Thm H	proved (ZFC)
(Q) Is full $\mathbf{Vec}$ admissible / F projective? (Question 1)	2026-09-02/04	OPEN (conj. indep. ZFC)

Outlook. The theory of vector-space containers is complete and clean on the finite-dimensional locus: the extension functor is neither full nor faithful (faithful only when the target has ≤ 1 shape or the source is empty) and is not injective on objects, the failure is exactly the non-extensivity of $\mathbf{Vec}$, comonoids are families of algebras, and $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ realises admissibility off the connected-unit pole. The single deep frontier is Question (Q) — whether *infinite-dimensional* linear resources can still absorb external shape — now sharpened to a precise, ZFC-reduced homological question ($\text{Ext}^1(F, -) \equiv 0$?) with both classical decision routes provably closed and independence of ZFC conjectured. Settling it is the natural next step and would either produce a genuinely new irreducible composition base or complete the absorptive-dichotomy classification.

Draft — work in progress. Prepared for Neil Ghani / external requester, 2026-09-05. Internal sources: proofs/2026-08-18-linear-containers-vec.md, proofs/2026-08-19-vec-comonoids-algebras.md, proofs/2026-09-11-collapse-defect-formula.md, proofs/2026-09-01-gap1-setxvec-proved.md, proofs/2026-09-02-vec-admissibility-rigidity.md, proofs/2026-09-04-Q-homological-reduction-and-target-dimension-dichotomy.md, lean/Containers/Containers/VecCollapse.lean. Grades per the corresponding registry files.